06. Adaptive Sliding Mode Control for Mechanical Systems

Chapter Objectives

This chapter introduces a kind of adaptive sliding mode control for mechanical systems, stability analysis and simulation examples are given

§1. Adaptive Sliding Mode Control for Mechanical Systems

1. System Description

The uncertain mechanical system is described as

$$\dot{x}_1 = x_2 \quad (6.1)$$

$$m\dot{x}_2 = u(t) + \Delta \quad (6.2)$$

where, x_1 : position

x_2 : velocity

m : unknown constant moment inertia

Δ : uncertainty including matched/unmatched disturbances

Let $\theta = m$, then, Eq.(6.2) can be rewritten as

$$\theta\dot{x}_2 = u + \Delta \quad (6.3)$$

Assumptions

$$\theta \in \Omega \triangleq \{\theta: 0 < \theta_{min} \le \theta \le \theta_{max}\} \quad (6.4)$$

$$|\Delta| \le D \quad (6.5)$$

§1. Adaptive Sliding Mode Control for Mechanical Systems

2. Design of Adaptive Sliding Mode Controller

Select the sliding variable as

$$\begin{cases} s = \dot{e} + ce = x_2 - q \\ q = \dot{x}_d - ce \end{cases} \quad (6.6)$$

where, $e = x_1 - x_d$ and $c > 0$ must be Hurwitz → $\dot{s} = \dot{x}_2 - \dot{q}$

The controller is selected as

$$u = u_a + u_{s_1} + u_{s_2} \quad (6.7)$$

where, $u_a = \hat{\theta}\dot{q}$ (6.8)

$$u_{s_1} = -k_s s, \qquad k_s > 0 \quad (6.9)$$

$$u_{s_2} = -\eta \text{sgn}(s), \quad \eta > 0 \quad (6.10)$$

$\hat{\theta}$: estimation of θ u_a: adaptive compensation

u_{s_1}: the feedback item u_{s_2}: the robustness item

§1. Adaptive Sliding Mode Control for Mechanical Systems

2. Design of Adaptive Sliding Mode Controller

Rewritten Eq.(6.7) as

$$u = u_a + u_{s_1} + u_{s_2} = \hat{\theta}\dot{q} - k_s s - \eta \text{sgn}(s)$$

Select the Lyapunov function as

$$V = \frac{1}{2}\theta s^2 + \frac{1}{2\gamma}\tilde{\theta}^2 \quad (6.11)$$

where, $\tilde{\theta} = \hat{\theta} - \theta, \gamma > 0$

Therefore, we get

$$\dot{V} = \theta s\dot{s} + \frac{1}{\gamma}\tilde{\theta}\dot{\tilde{\theta}} = s(\theta\dot{x}_2 - \theta\dot{q}) + \frac{1}{\gamma}\tilde{\theta}\dot{\hat{\theta}}$$

Select the adaptive law as

$$\dot{\hat{\theta}} = -\gamma\dot{q}s \quad (6.12)$$

$s = \dot{e} + ce = x_2 - q, \dot{s} = \dot{x}_2 - \dot{q}, \tilde{\theta} = \hat{\theta} - \theta, \dot{\tilde{\theta}} = \dot{\hat{\theta}}$

§1. Adaptive Sliding Mode Control for Mechanical Systems

2. Design of Adaptive Sliding Mode Controller

$\dot{x}_1 = x_2$

$\theta\dot{x}_2 = u + \Delta \qquad \theta \in \Omega \triangleq \{\theta: 0 < \theta_{min} \le \theta \le \theta_{max}\}, |\Delta| \le D \qquad \tilde{\theta} = \hat{\theta} - \theta$

$e = x_1 - x_d \qquad q = \dot{x}_d - ce, c > 0 \qquad s = \dot{e} + ce = x_2 - q \qquad \dot{s} = \dot{x}_2 - \dot{q}$

$V = \frac{1}{2}\theta s^2 + \frac{1}{2\gamma}\tilde{\theta}^2$

$\dot{V} = \theta s\dot{s} + \frac{1}{\gamma}\tilde{\theta}\dot{\tilde{\theta}}$

$= s(\theta\dot{x}_2 - \theta\dot{q}) + \frac{1}{\gamma}\tilde{\theta}\dot{\hat{\theta}} = s(u + \Delta - \theta\dot{q}) + \frac{1}{\gamma}\tilde{\theta}\dot{\hat{\theta}}$

$= s\left(\hat{\theta}\dot{q} - \hat{\theta}\dot{q} - k_s s + k_s s - \eta \text{ sgn}(s) + \eta \text{ sgn}(s) + u + \Delta - \theta\dot{q}\right) + \frac{1}{\gamma}\tilde{\theta}\dot{\hat{\theta}}$

$= s\left(\frac{1}{\gamma}\gamma\tilde{\theta}\dot{q} - \hat{\theta}\dot{q} + k_s s + \eta \text{ sgn}(s) + u - k_s s - \eta \text{ sgn}(s) + \Delta\right) + \frac{1}{\gamma}\tilde{\theta}\dot{\hat{\theta}}$

$= s(-\hat{\theta}\dot{q} + k_s s + \eta \text{ sgn}(s) + u - k_s s - \eta \text{ sgn}(s) + \Delta) + \frac{1}{\gamma}\tilde{\theta}(\dot{\hat{\theta}} - \gamma\dot{q}s)$

$= s(-k_s s - \eta \text{ sgn}(s) + \Delta)$

$= -k_s s^2 - \eta|s| + \Delta s < -k_s s^2 \le 0$

§1. Adaptive Sliding Mode Control for Mechanical Systems

2. Design of Adaptive Sliding Mode Controller

$$V = \frac{1}{2}\theta s^2 + \frac{1}{2\gamma}\tilde{\theta}^2 \quad (6.11)$$

$$\dot{V} = s(\theta\dot{x}_2 - \theta\dot{q}) + \frac{1}{\gamma}\tilde{\theta}\dot{\tilde{\theta}} = s(u + \Delta - \theta\dot{q}) + \frac{1}{\gamma}\tilde{\theta}\dot{\tilde{\theta}}$$
$$= s(\hat{\theta}\dot{q} - k_s s - \eta\,\mathrm{sgn}(s) + \Delta - \theta\dot{q}) + \frac{1}{\gamma}\tilde{\theta}(-\gamma\dot{q}s)$$
$$= s(\tilde{\theta}\dot{q} - k_s s - \eta\,\mathrm{sgn}(s) + \Delta) - \tilde{\theta}(\dot{q}s)$$
$$= -k_s s^2 - \eta|s| + \Delta s < -k_s s^2 \le 0$$

In order to avoid $\hat{\theta}$ which is too big to make $u(t)$ too big, rewrite (6.12) by using the discontinuous projection mapping as Proj

$$\dot{\hat{\theta}} = \mathrm{Proj}_{\hat{\theta}}(-\gamma\dot{q}s), \quad \mathrm{Proj}_{\hat{\theta}}(.) = \begin{cases} 0 & \text{if } \hat{\theta} = \theta_{max} \text{ and } \cdot > 0 \\ 0 & \text{if } \hat{\theta} = \theta_{min} \text{ and } \cdot < 0 \\ \cdot & \text{otherwise} \end{cases} \quad (6.13)$$

$$\dot{\hat{\theta}} = -\gamma\dot{q}s \quad (6.12)$$

§1. Adaptive Sliding Mode Control for Mechanical Systems

Summarize

$\dot{x}_1 = x_2$

$m\dot{x}_2 = u + \Delta$ or $\theta\dot{x}_2 = u + \Delta$, $\theta \in \Omega \stackrel{\text{def}}{=} \{\theta: 0 < \theta_{min} \le \theta \le \theta_{max}\}$, $|\Delta| \le D$

$e = x_1 - x_d$

$q = \dot{x}_d - ce$, $c > 0$

$s = \dot{e} + ce = x_2 - q$, $\dot{s} = \dot{x}_2 - \dot{q}$

$\dot{\hat{\theta}} = -\gamma\dot{q}s$ (6.12)

$\dot{\hat{\theta}} = \mathrm{Proj}_{\hat{\theta}}(-\gamma\dot{q}s)$, $\mathrm{Proj}_{\hat{\theta}}(.) = \begin{cases} 0 & \text{if } \hat{\theta} = \theta_{max} \text{ and } \cdot > 0 \\ 0 & \text{if } \hat{\theta} = \theta_{min} \text{ and } \cdot < 0 \\ \cdot & \text{otherwise} \end{cases}$ (6.13)

$u_a = \hat{\theta}\dot{q}$

$u_{s_1} = -k_s s$, $k_s > 0$

$u_{s_2} = -\eta\mathrm{sgn}(s)$, $\eta > 0$ ($\eta = D + \varepsilon$)

$u = u_a + u_{s_1} + u_{s_2}$

§1. Adaptive Sliding Mode Control for Mechanical Systems

3. Simulation Example (Ex_06.01.m)

The plant $\dot{x}_1 = x_2$

$m\dot{x}_2 = u(t) + \Delta$

where, $m \in [0.5,\ 1.5] \rightarrow m_n = 1$

$\Delta = 0.5x_2 + 1.5\mathrm{sgn}x_2$

The desired trajectory $\sin t$

The initial condition $\boldsymbol{x}_0 = [0.5\ \ 0]^T$

Choosing $c = 15, k_s = 15, \gamma = 500, \eta = D + 0.01 = 2.01$

Using adaptive sliding mode controller ($M = 1$)

PD controller ($M = 2$)

adaptive updated law (6.12) ($N = 1$)

discontinuous projection mapping (6.13) ($N = 2$)

§1. Adaptive Sliding Mode Control for Mechanical Systems

3. Simulation Example

$$\begin{cases} \dot{x}_1 = x_2 \\ \theta\dot{x}_2 = u + \Delta \end{cases}, \ \Delta = 0.5x_2 + 1.5\mathrm{sgn}x_2$$

$$\rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0.5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}u + \begin{bmatrix} 0 \\ 1 \end{bmatrix}d$$

$d = 1.5\mathrm{sgn}x_2$

$x_d = \sin t$

$e = x_1 - x_d$

$q = \dot{x}_d - ce$, $c > 0$

$s = \dot{e} + ce = x_2 - q$

$u = u_a + u_{s_1} + u_{s_2}$

$u_a = \hat{\theta}\dot{q}$ $\quad u_{s_1} = -k_s s, k_s > 0$ $\quad u_{s_2} = -\eta\mathrm{sgn}(s), \eta > 0$

→ Calculate the update laws $\hat{\theta}$

§1. Adaptive Sliding Mode Control for Mechanical Systems

3. Simulation Example

$u = u_a + u_{s_1} + u_{s_2}$

- $u_a = \hat{\theta}\dot{q}$, with the update laws $\hat{\theta}$
- $u_{s_1} = -k_s s$, $k_s > 0$
- $u_{s_2} = -\eta\mathrm{sgn}(s)$, $\eta > 0$

$M = 1$ using adaptive sliding mode controller

$M = 2$ using PD controller

$N = 1$ using updated law (6.12)

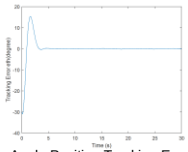
$N = 2$ using discontinuous projection mapping (6.13)

$$\dot{\hat{\theta}} = -\gamma\dot{q}s \quad (6.12)$$

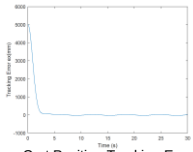
$$\dot{\hat{\theta}} = \mathrm{Proj}_{\hat{\theta}}(-\gamma\dot{q}s), \quad \mathrm{Proj}_{\hat{\theta}}(.) = \begin{cases} 0 & \text{if } \hat{\theta} = \theta_{max} \text{ and } \cdot > 0 \\ 0 & \text{if } \hat{\theta} = \theta_{min} \text{ and } \cdot < 0 \\ \cdot & \text{otherwise} \end{cases} \quad (6.13)$$

§1. Adaptive Sliding Mode Control for Mechanical Systems

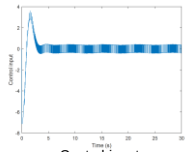
3. Simulation Example $M = 1, N = 1$



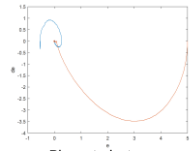
Angle Position Tracking Error



Cart Position Tracking Error



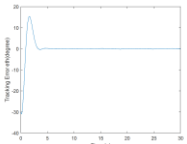
Control input



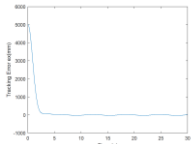
Phase trajectory

§1. Adaptive Sliding Mode Control for Mechanical Systems

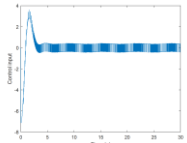
3. Simulation Example $M = 1, N = 2$



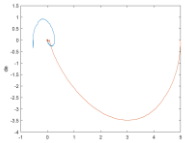
Angle Position Tracking Error



Cart Position Tracking Error



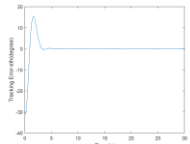
Control input



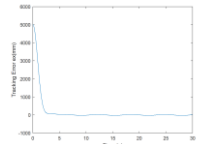
Phase trajectory

§1. Adaptive Sliding Mode Control for Mechanical Systems

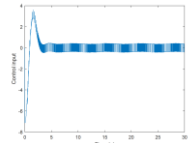
3. Simulation Example $M = 2, N = 1$



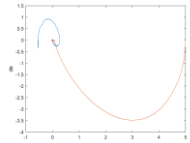
Angle Position Tracking Error



Cart Position Tracking Error



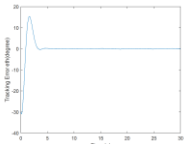
Control input



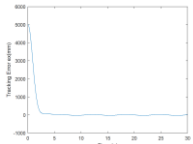
Phase trajectory

§1. Adaptive Sliding Mode Control for Mechanical Systems

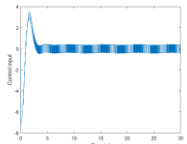
3. Simulation Example $M = 2, N = 2$



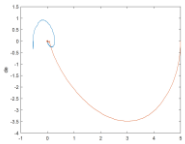
Angle Position Tracking Error



Cart Position Tracking Error



Control input



Phase trajectory

§2. Adaptive Sliding Mode Control of Inverted Pendulum

1. System Description

The kinetic equation of the inverted pendulum is given as follow

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(\boldsymbol{x}) + g(\boldsymbol{x})u \end{cases}$$

$$f(\boldsymbol{x}) = \frac{g\sin x_1 - m_p l x_2^2 \cos x_1 \sin x_1/(m_c + m_p)}{l\left[4/3 - m_p \cos^2 x_1/(m_c + m_p)\right]}$$

$$g(\boldsymbol{x}) = \frac{\cos x_1/(m_c + m_p)}{l\left[4/3 - m_p \cos^2 x_1/(m_c + m_p)\right]}$$

x_1 : the pendulum angle θ x_2 : the pendulum speed $\dot{\theta}$

g : gravity acceleration, $g = 9.81 m/s^2$

m_c : cart mass, $m_c = 0.5kg$ m_p: pendulum mass, $m_p = 0.5kg$

$2l$: pendulum length, $l = 0.3m$ u : the control input

Desired trajectory $\theta_d(t) = 0.1\sin t$; initial state $[0.01 \quad 0]$

§2. Adaptive Sliding Mode Control of Inverted Pendulum

2. Control System Design

Arrange Eq. (6.14)

$$l\left[4/3(m_c + m_p) - m_p \cos^2 x_1\right]\dot{x}_2 = g\sin x_1 (m_c + m_p) - m_p l x_2^2 \cos x_1 \sin x_1 + \cos x_1 u$$

$$\rightarrow l\left[4/3(m_c + m_p)\frac{1}{\cos x_1} - m_p \cos x_1\right]\dot{x}_2 = g\tan x_1 (m_c + m_p) - m_p l x_2^2 \sin x_1 + u$$

Since $I = \frac{1}{3} m_p l^2 \rightarrow \frac{4}{3} l = \frac{I + m_p l^2}{m_p l}$, and

$$\left[\frac{I + m_p l^2}{m_p l}(m_c + m_p)\frac{1}{\cos x_1} - m_p l \cos x_1\right]\dot{x}_2 = g\tan x_1 (m_c + m_p) - m_p l x_2^2 \sin x_1 + u \quad (6.15)$$

§2. Adaptive Sliding Mode Control of Inverted Pendulum

2. Control System Design

In order to design controller without need model information, choose

$$\phi_1 = (m_c + m_p)\frac{I + m_p l^2}{m_p l},\ \phi_2 = (m_c + m_p)g,\ \phi_3 = m_p l$$

Considering outer disturbance, transfer (6.15) to state equation as

$$\begin{cases} \dot{x}_1 = x_2 \\ g(x_1)\dot{x}_2 = u + \phi_2 \tan x_1 - \phi_3 x_2^2 \sin x_1 - dt \end{cases} \quad (6.16)$$

where, $g(x_1) = \phi_1(1/\cos x_1) - \phi_3 \cos x_1$, dt: disturbance

Define θ_d as the ideal position signal, and

$e = x_1 - \theta_d$ as the position tracking error

The sliding function is $s = ce + \dot{e}$, $c > 0$

$$\left[\frac{I + m_p l^2}{m_p l}(m_c + m_p)\frac{1}{\cos x_1} - m_p l \cos x_1\right]\dot{x}_2 = g\tan x_1 (m_c + m_p) - m_p l x_2^2 \sin x_1 + u \quad (6.15)$$

§2. Adaptive Sliding Mode Control of Inverted Pendulum

2. Control System Design

The Lyapunov function is defined as

$$V = V_1 + V_2$$

where, $V_1 = \frac{1}{2} g(x_1) s^2$

$$V_2 = \frac{1}{2\gamma_1}(\phi_1 - \hat{\phi}_1)^2 + \frac{1}{2\gamma_2}(\phi_2 - \hat{\phi}_2)^2 + \frac{1}{2\gamma_3}(\phi_3 - \hat{\phi}_3)^2$$

with, $\gamma_i > 0$, $\hat{\phi}_i$ is estimation value of ϕ_i, $i = 1,2,3, \ldots$

§2. Adaptive Sliding Mode Control of Inverted Pendulum

2. Control System Design

$$V_1 = \frac{1}{2} g(x_1) s^2$$

$$\rightarrow \dot{V}_1 = \frac{1}{2}\dot{g}(x_1)s^2 + g(x_1)s\dot{s}$$

$$V_2 = \frac{1}{2\gamma_1}(\phi_1 - \hat{\phi}_1)^2 + \frac{1}{2\gamma_2}(\phi_2 - \hat{\phi}_2)^2 + \frac{1}{2\gamma_3}(\phi_3 - \hat{\phi}_3)^2$$

$$\rightarrow \dot{V}_2 = -\frac{1}{\gamma_1}(\phi_1 - \hat{\phi}_1)\dot{\hat{\phi}}_1 - \frac{1}{\gamma_2}(\phi_2 - \hat{\phi}_2)\dot{\hat{\phi}}_2 - \frac{1}{\gamma_3}(\phi_3 - \hat{\phi}_3)\dot{\hat{\phi}}_3$$

where, $g(x_1)s\dot{s} = g(x_1)s(c\dot{e} + \dot{x}_2 - \ddot{\theta}_d)$

$$= s[g(x_1)(c\dot{e} - \ddot{\theta}_d)] + [u + \phi_2 \tan x_1 - \phi_3 x_2^2 \sin x_1 - dt]$$

$$\dot{g}(x_1) = [\phi_1(1/\cos x_1)\tan x_1 + \phi_3 \sin x_1]x_2$$

$g(x_1) = \phi_1(1/\cos x_1) - \phi_3 \cos x_1$

§2. Adaptive Sliding Mode Control of Inverted Pendulum

2. Control System Design

$$\dot{V}_1 = \frac{1}{2}\dot{g}s^2 + gs\dot{s}$$

$$= \frac{1}{2}\left[\phi_1\left(\frac{1}{\cos x_1}\right)\tan x_1 + \phi_3 \sin x_1\right]x_2 s^2 + s\left\{\left[\phi_1\left(\frac{1}{\cos x_1}\right) - \phi_3 \cos x_1\right](c\dot{e} - \ddot{\theta}_d)\right\} + [u + \phi_2 \tan x_1 - \phi_3 x_2^2 \sin x_1 - dt]$$

$$= \phi_1\left[\frac{1}{2}x_2 s^2\left(\frac{1}{\cos x_1}\right)\tan x_1 + s\left(\frac{1}{\cos x_1}\right)(c\dot{e} - \ddot{\theta}_d)\right] + \phi_1 s \tan x_1 + \phi_3\left[\frac{1}{2}x_2 s^2 \sin x_1 - s x_2^2 \sin x_1 - s\cos x_1(c\dot{e} - \ddot{\theta}_d)\right] + s(u - dt)$$

$g(x_1) = \phi_1(1/\cos x_1) - \phi_3 \cos x_1 \qquad \dot{g}(x_1) = [\phi_1(1/\cos x_1)\tan x_1 + \phi_3 \sin x_1]x_2$

$g(x_1)s\dot{s} = s[g(x_1)(c\dot{e} - \ddot{\theta}_d))] + [u + \phi_2 \tan x_1 - \phi_3 x_2^2 \sin x_1 - dt]$

§2. Adaptive Sliding Mode Control of Inverted Pendulum

2. Control System Design

The control law is defined as (η: constant, $\eta \geq \max|dt|$)

$$u = -\eta\,\mathrm{sgn}(s) - \hat{\phi}_1\left[\frac{1}{2}s x_2 \frac{1}{\cos x_1}\tan x_1 + \frac{1}{\cos x_1}(c\dot{e} - \ddot{\theta}_d)\right] - \hat{\phi}_2 \tan x_1 - \hat{\phi}_3\left[\frac{1}{2}s x_2 \sin x_1 - x_2^2 \sin x_1 - \cos x_1(c\dot{e} - \ddot{\theta}_d)\right] \quad (6.18)$$

Substitute the control law (6.18) to $\dot{V}_1$, we have

$$\dot{V}_1 = -\eta|s| - sdt + (\phi_1 - \hat{\phi}_1)\left[\frac{1}{2}s^2 x_2 \frac{1}{\cos x_1}\tan x_1 + s\frac{1}{\cos x_1}(c\dot{e} - \ddot{\theta}_d)\right] + (\phi_2 - \hat{\phi}_2)(s \tan x_1) + (\phi_3 - \hat{\phi}_3)\left[(1/2)s^2 x_2 \sin x_1 - s x_2^2 \sin x_1 - s\cos x_1(c\dot{e} - \ddot{\theta}_d)\right]$$

$\dot{V}_1 = \phi_1[(1/2)x_2 s^2(1/\cos x_1)\tan x_1 + s(1/\cos x_1)(c\dot{e} - \ddot{\theta}_d)] + \phi_1 s \tan x_1$
$+\phi_3[(1/2)x_2 s^2 \sin x_1 - s x_2^2 \sin x_1 - s\cos x_1(c\dot{e} - \ddot{\theta}_d)] + s(u - dt)$

§2. Adaptive Sliding Mode Control of Inverted Pendulum

2. Control System Design

$$\dot{V} = \dot{V}_1 + \dot{V}_2$$

$$= -\eta|s| - sdt + (\phi_1 - \hat{\phi}_1)\left\{\left[\frac{1}{2}s^2 x_2 \frac{1}{\cos x_1}\tan x_1 + s\frac{1}{\cos x_1}(c\dot{e} - \ddot{\theta}_d)\right] - \frac{1}{\gamma_1}\dot{\hat{\phi}}_1\right\} + (\phi_2 - \hat{\phi}_2)\left(s \tan x_1 - \frac{1}{\gamma_2}\dot{\hat{\phi}}_2\right) + (\phi_3 - \hat{\phi}_3)\left\{\left[\frac{1}{2}s^2 x_2 \sin x_1 - s x_2^2 \sin x_1 - s\cos x_1(c\dot{e} - \ddot{\theta}_d)\right] - \frac{1}{\gamma_3}\dot{\hat{\phi}}_3\right\}$$

$\dot{V}_1 = -\eta|s| - sdt + (\phi_1 - \hat{\phi}_1)\left[\frac{1}{2}s^2 x_2 \frac{1}{\cos x_1}\tan x_1 + s\frac{1}{\cos x_1}(c\dot{e} - \ddot{\theta}_d)\right]$
$+(\phi_2 - \hat{\phi}_2)(s \tan x_1) + (\phi_3 - \hat{\phi}_3)\left[(1/2)s^2 x_2 \sin x_1 - s x_2^2 \sin x_1 - s\cos x_1(c\dot{e} - \ddot{\theta}_d)\right]$

$\dot{V}_2 = -\frac{1}{\gamma_1}(\phi_1 - \hat{\phi}_1)\dot{\hat{\phi}}_1 - \frac{1}{\gamma_2}(\phi_2 - \hat{\phi}_2)\dot{\hat{\phi}}_2 - \frac{1}{\gamma_3}(\phi_3 - \hat{\phi}_3)\dot{\hat{\phi}}_3$

§2. Adaptive Sliding Mode Control of Inverted Pendulum

2. Control System Design

$$\dot{V} = -\eta|s| - sdt + (\phi_1 - \hat{\phi}_1)\underbrace{\left\{\left[\frac{1}{2}s^2 x_2 \frac{1}{\cos x_1}\tan x_1 + s\frac{1}{\cos x_1}(c\dot{e} - \ddot{\theta}_d)\right] - \frac{1}{\gamma_1}\dot{\hat{\phi}}_1\right\}}_{=0} + (\phi_2 - \hat{\phi}_2)\underbrace{\left(s \tan x_1 - \frac{1}{\gamma_2}\dot{\hat{\phi}}_2\right)}_{=0} + (\phi_3 - \hat{\phi}_3)\underbrace{\left\{\left[\frac{1}{2}s^2 x_2 \sin x_1 - s x_2^2 \sin x_1 - s\cos x_1(c\dot{e} - \ddot{\theta}_d)\right] - \frac{1}{\gamma_3}\dot{\hat{\phi}}_3\right\}}_{=0}$$

The adaptive law is designed as

$$\dot{\hat{\phi}}_1 = \gamma_1\left[\frac{1}{2}s^2 x_2 \frac{1}{\cos x_1}\tan x_1 + s\frac{1}{\cos x_1}(c\dot{e} - \ddot{\theta}_d)\right]$$

$$\dot{\hat{\phi}}_2 = \gamma_2 s \tan x_1 \quad (6.19)$$

$$\dot{\hat{\phi}}_3 = \gamma_3\left[\frac{1}{2}s^2 x_2 \sin x_1 - s x_2^2 \sin x_1 - s\cos x_1(c\dot{e} - \ddot{\theta}_d)\right]$$

Then $\dot{V} = -\eta|s| - sdt \leq 0$

§2. Adaptive Sliding Mode Control of Inverted Pendulum

3. Simulation Example (Ex_06.02.m)

The kinetic equation of the inverted pendulum is given as follow

$$\begin{cases}\dot{x}_1 = x_2 \\ \dot{x}_2 = f(\boldsymbol{x}) + g(\boldsymbol{x})u\end{cases}$$

$$f(\boldsymbol{x}) = \frac{g\sin x_1 - m_p l x_2^2 \cos x_1 \sin x_1/(m_c + m_p)}{l\left[4/3 - m_p \cos^2 x_1/(m_c + m_p)\right]}$$

$$g(\boldsymbol{x}) = \frac{\cos x_1/(m_c + m_p)}{l\left[4/3 - m_p \cos^2 x_1/(m_c + m_p)\right]}$$

x_1 : the pendulum angle θ x_2 : the pendulum speed $\dot{\theta}$

g : gravity acceleration, $g = 9.81 m/s^2$

m_c : cart mass, $m_c = 0.5kg$ m_p: pendulum mass, $m_p = 0.5kg$

$2l$: pendulum length, $l = 0.3m$ u : the control input

Desired trajectory $\theta_d(t) = 0.1\sin t$; initial state $[0.01 \quad 0]$

§2. Adaptive Sliding Mode Control of Inverted Pendulum

3. Simulation Example

$$\begin{bmatrix}\dot{x}_1 \\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}x_2 \\ f(\boldsymbol{x})\end{bmatrix} + \begin{bmatrix}0 \\ g(\boldsymbol{x})\end{bmatrix} u + \begin{bmatrix}0 \\ 1\end{bmatrix} dt$$

$d = \sin t$; $\phi_1(0) = 0$; $\phi_2(0) = 0$; $\phi_3(0) = 0$

$c = 10$, $\eta = 0.05$, $\gamma_1 = \gamma_2 = \gamma_3 = 150$; $e = x_1 - \theta_d$; $s = ce + \dot{e}$

$$\dot{\hat{\phi}}_1 = \gamma_1 \left[\frac{1}{2} s^2 x_2 \frac{1}{\cos x_1} \tan x_1 + s \frac{1}{\cos x_1}(c\dot{e} - \ddot{\theta}_d)\right]$$

$$\dot{\hat{\phi}}_2 = \gamma_2 \tan x_1$$

$$\dot{\hat{\phi}}_3 = \gamma_3 \left[0.5 s^2 x_2 \sin x_1 - s x_2^2 \sin x_1 - s \cos x_1 (c\dot{e} - \ddot{\theta}_d)\right]$$

$$u = -\eta\, \mathrm{sgn}(s) - \hat{\phi}_1 \left[\frac{1}{2} s x_2 \frac{1}{\cos x_1} \tan x_1 + \frac{1}{\cos x_1}(c\dot{e} - \ddot{\theta}_d)\right]$$
$$-\hat{\phi}_2 \tan x_1 - \hat{\phi}_3 \left[0.5 s x_2 \sin x_1 - x_2^2 \sin x_1 - \cos x_1 (c\dot{e} - \ddot{\theta}_d)\right]$$